SmartLogix AI

AI-Powered Intelligent Logistics & Autonomous Delivery Platform

An intelligent platform that leverages Machine Learning, Computer Vision, LLMs, and AI Agents to optimize multi-modal logistics, monitor fleet health, and deliver seamless customer experiences.

Machine Learning | Computer Vision | LLM & NLP | AI Agents | Cloud (AWS)



- Intelligent Logistics Decisioning
- Optimized Routes & Fleet Allocation
- Predictive Maintenance & Health Monitoring
- AI-Powered Chatbot & Customer Support
- Real-time Alerts & Automated Notifications

HCL GUVI Capstone Project

Domain: Logistics & Supply Chain

Technologies: ML | DL | CV | NLP | LLM | Agents | AWS

SmartLogix AI
AI-Powered Intelligent Logistics Platform

Problem Statement & Our Solution

HCL | GUVI
Skill Up. Level Up

CURRENT LOGISTICS CHALLENGES

Traditional logistics systems face multiple inefficiencies leading to higher cost and poor customer experience.



Wrong Transport Selection
No intelligent system to choose the best mode (drone, bike, truck, ship, etc.) based on real-time factors.

Delayed Deliveries
Traffic, weather and route inefficiencies lead to inaccurate ETAs and late deliveries.

High Operational Cost
Inefficient routing, underutilized fleet and high fuel/energy consumption.

No Predictive Maintenance
Unexpected vehicle/drone failures increase downtime and repair cost.

Limited Customer Support
Customers lack real-time visibility and intelligent support for their orders.

VS

SMARTLOGIX AI – OUR SOLUTION

An AI-powered platform that intelligently decides, optimizes, monitors and automates the entire delivery lifecycle.



KEY BENEFITS

Faster Deliveries
Accurate ETA prediction & optimized routing

Lower Cost
Reduced operational and fuel cost

Higher Reliability
Predictive maintenance reduces breakdowns

Better Utilization
Efficient fleet & drone usage

Intelligent Support
AI chatbot & agents for seamless experience

Sustainable Operations
Eco-friendly deliveries with optimized resources

AI TECHNOLOGIES WE LEVERAGE

Machine Learning | Computer Vision | NLP | LLMs & RAG | AI Agents | Cloud Computing

Our Goal: Deliver the **right package**, through the **right mode**, on the **right route**, at the **right time** – every time.

Smarter Logistics. Better Tomorrow.

SmartLogix AI

AI-Powered Intelligent Logistics & Autonomous Delivery Platform

End-to-End System Workflow

From Order to Intelligent Delivery & Customer Delight

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01 Customer Places Order



Order is placed on the platform with delivery details.

02 Order Management



Order is validated and stored in the system.

03 ML Logistics Classification



ML model predicts the most suitable delivery mode (Drone/Bike/Van/Truck/Ship).

04 Route Optimization



Best route is optimized based on distance, traffic, battery, payload & constraints.

05 Fleet Allocation



Optimal vehicle or drone is assigned from available fleet.

06 ETA Prediction



Estimated delivery time is predicted using regression model.

07 Delivery Execution



Delivery is executed and real-time location is tracked.

08 Real-time Monitoring



Live tracking of drone/vehicle, location, speed, and status.

09 Drone Inspection (CV)



Computer Vision model detects drone damages from inspection images.

10 Predictive Maintenance



ML model predicts health status and maintenance requirements.

11 Data Storage & Logs



All data, logs, telemetry and transactions are stored securely.

12 AI Agents & LLM Chatbot



RAG-based LLM chatbot & AI Agents assist customers and operations.

DATA & AI LAYER

Input Data Sources
Orders, Customers, Products, Weather, Traffic, Maps, Fleet, Sensor Data, Reviews

Machine Learning Models
Classification, Regression, Predictive Maintenance, Optimization Models

Computer Vision Model
YOLO-based model for drone damage detection and inspection

NLP & LLM (RAG)
Retrieval Augmented Generation over DB, docs, logs & policies

AI Agents
Support Agent, Logistics Agent, Recommendation Agent, Maintenance Agent, Notification Agent

Database (RDS)
PostgreSQL storing orders, customers, fleet, logs, reviews, maintenance and more.

Cloud Services (AWS)
EC2, RDS, S3, Lambda, SES, CloudWatch, IAM for secure & scalable infrastructure

OUTPUT & ACTIONS

Best Delivery Mode Selected

Optimized Route with ETA

Real-time Tracking & Updates

Automated Notifications

AI Chatbot Assistance

Insights & Analytics Dashboard

Customer Satisfaction

Outcome: Smarter Decisions • Optimized Deliveries • Lower Costs • Happy Customers

HCL GUVI Capstone Project

SmartLogix AI
Smarter Decisions. Optimized Deliveries.

AI & Machine Learning Modules

Intelligent Decisions at Every Step

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01 Intelligent Logistics Classification

Predicts the most suitable delivery mode based on multiple real-time and static factors.

INPUT FEATURES

- Distance
- Package Weight
- Package Dimensions
- Weather Conditions
- Traffic Level
- Customer Priority
- Fragility
- Delivery Deadline
- Shipping Cost
- Seller Location
- Vehicle Availability
- Historical Data

MODEL PIPELINE

Processed Data → Feature Engineering → ML Model (Random Forest / XGBoost) → Drone, Bike, Van, Truck, Air Cargo, Ship

EXPECTED PERFORMANCE

Accuracy	F1 Score	Precision	Recall
≥ 88%	≥ 0.88	≥ 0.88	≥ 0.88

02 ETA Prediction

Predicts the Estimated Delivery Time for accurate customer commitment.

KEY FEATURES

- Distance
- Traffic
- Weather
- Delivery Mode
- Time of Day
- Driver Rating
- Vehicle Speed
- Package Weight
- Historical ETA

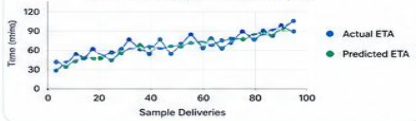
MODEL PIPELINE

Historical Data → Feature Engineering → Regression Model (XGBoost / RF) → Predicted ETA

EXPECTED PERFORMANCE

MAE	RMSE	R^2 Score	MAPE
≤ 15 minutes	≤ 20 minutes	≥ 0.85	≤ 12%

Predicted vs Actual ETA (Example)



03 Route Optimization

Finds the most efficient delivery route considering constraints and objectives.

CONSIDERATIONS

- Shortest Distance
- Minimum Delivery Time
- Traffic Conditions
- Vehicle Capacity
- Battery Constraints (Drone)
- Delivery Priority
- Time Windows
- Payload Optimization
- Return Trip Feasibility

OPTIMIZATION APPROACH

Delivery Locations → Constraints & Objectives → Optimization Engine (OR-Tools / GA) → Optimized Route

EXPECTED OUTCOME

Distance Reduction	Time Reduction	Cost Reduction	Fleet Utilization
≥ 12%	≥ 15%	≥ 10%	≥ 75%

Route Optimization (Example)



04 Predictive Maintenance

Predicts potential failures and maintenance needs to prevent breakdowns and ensure reliability.

SENSOR & VEHICLE DATA

- Battery Health
- Motor Temperature
- Motor RPM
- Vibration Level
- Flight / Running Hours
- Payload Weight
- Crash History
- Environmental Factors

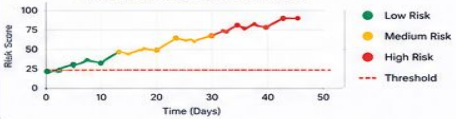
MODEL PIPELINE

Sensor Data → Data Preprocessing → ML Model (RF / XGBoost / LSTM) → Healthy, Service Required, Critical

EXPECTED PERFORMANCE

Accuracy	Recall	F1 Score	False Alarm Rate
≥ 90%	≥ 0.90	≥ 0.90	≤ 10%

Maintenance Risk Trend (Example)



KEY TAKEAWAY

Our AI/ML models work together to ensure the right mode, the right route, the right time and the right health for every delivery.

- Data-Driven Decisions
- Optimized Operations
- Proactive Maintenance
- Happier Customers

SmartLogix AI

AI-Powered Intelligent Logistics Platform

CV, NLP, LLM & AI AGENTS – Intelligent Ecosystem

Perception, Understanding, Reasoning & Action for Smarter Logistics

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1 COMPUTER VISION

Drone Damage Detection using YOLOv11

Drone Image Capture → YOLOv11 Model → Damage Detection → Inspection Report

DETECTION EXAMPLE



- Healthy (No Damage)
- Broken Propeller
- Motor Damage
- Frame Crack
- Battery Damage
- Landing Gear Damage
- Camera Damage

MODEL PERFORMANCE

mAP@50	mAP@50-95	Precision	Recall	F1 Score
91.2%	72.4%	93.1%	90.0%	91.5%

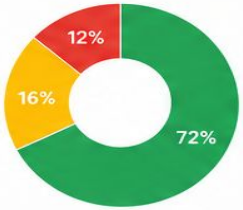
Automated visual inspection ensures safe and reliable drone operations with early damage detection.

2 NLP – SENTIMENT ANALYSIS

Understand Customer Feedback

Customer Reviews → Text Preprocessing → NLP Model (BERT) → Sentiment Classification

SENTIMENT DISTRIBUTION



- **Positive (72%)** Happy with fast delivery and product quality.
- **Neutral (16%)** Average experience, improvement possible.
- **Negative (12%)** Delayed delivery and package issues.

SAMPLE REVIEW ANALYSIS

Great service!	Average experience	Not satisfied
Package arrived on time and in perfect condition.	Delivery was okay, but packaging can be improved.	Delivery delayed and product received was damaged.
Positive	Neutral	Negative

NLP helps in understanding customer sentiments and drives data-backed improvements.

3 LLM + RAG + AI AGENTS

Intelligent Conversations. Real-time Decisions.

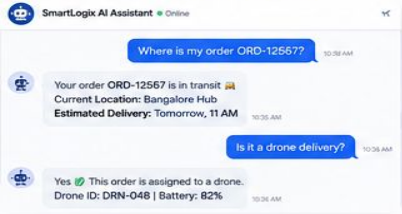
RAG ARCHITECTURE

Data Sources (Orders, Products, FAQs, Policies, Reviews, Logs) → Document Loader → Text Chunking & Embeddings → Vector DB (FAISS) → Retriever → LLM (Answer)

AI AGENTS ECOSYSTEM

Support Agent	Logistics Agent	Recommendation Agent	Maintenance Agent	Notification Agent
Handles order tracking, issues & FAQs	Provides delivery updates, ETA & routing info	Recommends products based on preferences	Monitors fleet health & raises service alerts	Sends alerts via Email, SMS, WhatsApp

CHATBOT DEMO



WHAT IT CAN DO

- Track Orders & Shipments
- Answer FAQs
- Provide Delivery ETAs
- Compare Products
- Analyze Reviews
- Raise Service Requests
- Smart Recommendations

RAG + LLM + AI Agents deliver accurate, context-aware answers and proactive operational support.

AI ECOSYSTEM BENEFITS

- **See Clearly** — Computer Vision ensures visual intelligence & safety assurance
- **Understand Deeply** — NLP extracts insights from unstructured data & customer feedback
- **Think Intelligently** — LLMs understand context and generate human-like responses
- **Act Proactively** — AI Agents take intelligent actions & automate business operations
- **Business Impact** — Better experience, lower costs, higher efficiency & smarter decisions